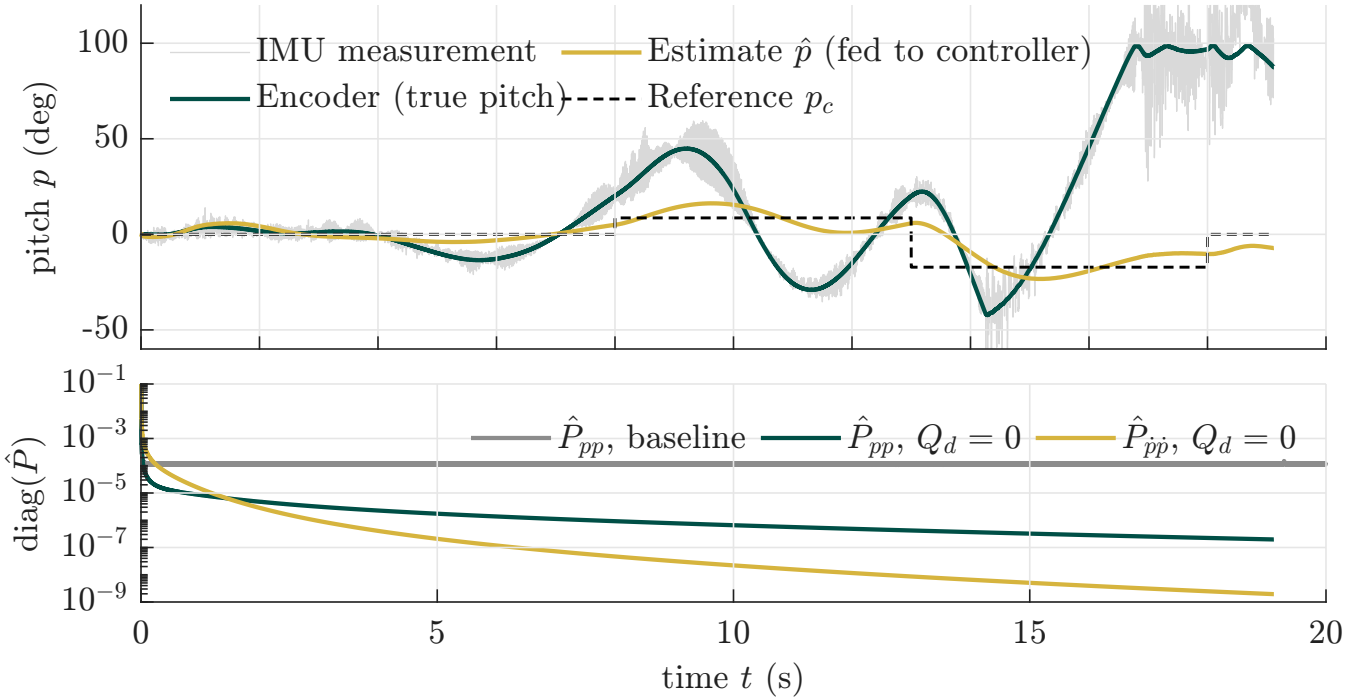

IMU measurement
Estimate $\hat{p}$ (fed to controller)
Encoder (true pitch)
Reference p_c
pitch p (deg)
100
50
0
-50
$\hat{P}_{pp}$, baseline
$\hat{P}_{pp}$, $Q_d = 0$
$\hat{P}_{\dot{p}\dot{p}}$, $Q_d = 0$
diag($\hat{P}$)
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
0
5
10
15
20
time t (s)